EXTERNAL HALL / ABI: 3.3V or 5V logic -> 3.3V Schmitt buffer -> reset isolation

Joint feedback supply: fixed, protected 5V shared with J502 fan.
J601.1 = 5V output; J601.2/6 = GND. No power input.
For a 5V Hall/ABI encoder; no voltage selector.
Sensor + pull-ups target <=25mA; shared accessory target <=300mA.

D601 rail isolated from supplies.
C604 is required; no 3V3 clamp tie.

INPUT OPTIONS
Default: 3.3V / 5V push-pull; pull-ups DNP.
Open-drain: fit R608-R610 (4.7k to protected 5V).
Do not fit pull-ups for externally driven push-pull.
100k pull-downs define disconnected inputs.
Single-ended only; common ground required.

D601
TPD3E001DRYR
C604
100nF 50V

J601
EXT ENCODER / HALL

U604
TPS2553DBVR
R604
10k
R606
49.9k 1%
C606
4.7uF

U604: R618=49.9k 1% =>520mA nominal shared limit.
EXT_VCC feeds J601 sensor + J502 fan + U503 PWM buffer.
EXT_POWER_FAULT_N -> PC13; shared branch fault.
Stop torque on fault or external-feedback loss; verify fan startup margin.

R608
4.7k DNP
R609
4.7k DNP
R610
4.7k DNP
R605
100R
R606
100R
R607
100R
R615
100k
R616
100k
R617
100k
U603A
U603C
U603B
74LVC3G17

U602A
TMUX1574PW
TP1016
FEEDBACK_B_H2
TP1015
FEEDBACK_A_H1
FEEDBACK_I_H3

U602B
TMUX1574PW

U603D
74LVC3G17

C607
100nF 16V X7R
C605
100nF
C603
100nF

R305
10k
R306
10k
R307
10k

U602C
TMUX1574PW

Reset: mux disabled, PB8/BOOT0 held low by R307.
EN_N=0 enables VM sensing; wait >=2ms before ADC read.
SEL=0 grounds TIM4 inputs; SEL=1 connects external Hall/ABI.
Onboard absolute-angle SPI is independent of this switch.

3.3V / 5V external inputs; interface revision
ShiroFOC
Sheet: /
File: 09_Feedback_Selection.kicad_sch
Title: External Hall / ABI conditioning and reset isolation
Size: A3 | Date: 2026-09-26 | **Rev: A1-Drawing**
KiCad E.D.A. 9.0.4 | Id: 1/1